EIN 4343 Inventory and Supply Chain Systems
Spring 26: Homework Assignment 2: Part 3
Solutions

1. **[3 points]** Consider a product for which the fixed ordering cost is $500 per order. The annual demand for the product is relatively level and is 10,000 units per year. The system uses an annual inventory carrying rate of 25 percent.
   - **a.** Suppose that an <u>all-units discount scheme</u> is effect with $c_0 = 5$, $c_1 = 4$, and c_2=3. What are the EOQ values that correspond to each of these price levels?
   - **b.** What is the optimal solution if the breakpoints are b_0=0, b_1=3,000 and b_2=6,000?
   - **c.** What is the optimal solution if the breakpoints are b_0=0, b_1=5,000 and b_2=10,000?

SOLUTION:

a. We have the following:

$$q_0^{EOQ} = \sqrt{\frac{(2)(500)(10{,}000)}{(0.25)(5)}} = 2{,}828.43 \text{ units}$$

$$q_1^{EOQ} = \sqrt{\frac{(2)(500)(10{,}000)}{(0.25)(4)}} = 3{,}162.28 \text{ units}$$

$$q_2^{EOQ} = \sqrt{\frac{(2)(500)(10{,}000)}{(0.25)(3)}} = 3{,}651.49 \text{ units}$$

b. If the breakpoints are {0, 3,000, 6,000}, we have:

$q_0^{EOQ} = 2{,}828.43$ units; Compare against the range $0 \leq q < 3{,}000$;
Case 2 applies (between the lower and upper bounds);
$q_0^* = 2{,}828.43$ units.

$q_1^{EOQ} = 3{,}162.28$ units; Compare against the range $3{,}000 \leq q < 6{,}000$;
Case 2 applies (between the lower and upper bounds);
$q_1^* = 3{,}162.28$ units.

$q_2^{EOQ} = 3{,}651.48$ units; Compare against the range $6{,}000 \leq q < \infty$;
Case 1 applies (less than the lower bound);
$q_2^* = 6{,}000$ units

Hence, we have 3 candidate solutions to consider:

$$C(2{,}828.43{,}5) = (5)(10{,}000) + \frac{(500)(10{,}000)}{2{,}828.43} + \frac{(0.25)(5)(2{,}828.43)}{2}$$
$$= 50{,}000 + 1{,}767.77 + 1{,}767.77 = \$53{,}535.54/\text{year}$$

$$C(3{,}162.28{,}4) = (4)(10{,}000) + \frac{(500)(10{,}000)}{3{,}162.28} + \frac{(0.25)(4)(3{,}000)}{2}$$
$$= 40{,}000 + 1{,}581.14 + 1{,}581.14 = \$43{,}162.28/\text{year}$$

$$C(6{,}000{,}3) = (3)(10{,}000) + \frac{(500)(10{,}000)}{6{,}000} + \frac{(0.25)(3)(6{,}000)}{2}$$
$$= 30{,}000 + 833.33 + 2{,}250 = \$33{,}083.33/\text{year}$$

The optimal solution is to order 6,000 units at a unit cost of $3/unit.

c. If the breakpoints are {0, 5,000, 10,000}, we have:

$q_0^{EOQ} = 2{,}828.43$ units; Compare against the range $0 \leq q < 5{,}000$;
Case 2 applies (between the lower and upper bounds);
$q_0^* = 2{,}828.43$ units.

$q_1^{EOQ} = 3{,}162.28$ units; Compare against the range $5{,}000 \leq q < 10{,}000$;
Case 1 applies (less than the lower bound);
$q_1^* = 5{,}000$ units.

$q_2^{EOQ} = 3{,}651.49$ units; Compare against the range $10{,}000 \leq q < \infty$;
Case 1 applies (less than the lower bound);
$q_2^* = 10{,}000$ units

Hence, we have 3 candidate solutions to consider:

$$C(2{,}828.43{,}5) = (5)(10{,}000) + \frac{(500)(10{,}000)}{2{,}828.43} + \frac{(0.25)(5)(2{,}828.43)}{2} = \$53{,}535.54/\text{year}$$

$$C(5{,}000,4) = (4)(10{,}000) + \frac{(500)(10{,}000)}{5{,}000} + \frac{(0.25)(4)(5{,}000)}{2} = 40{,}000 + 1{,}000 + 2{,}500$$
$$= \$43{,}500/\text{year}$$

$$C(10{,}000,3) = (3)(10{,}000) + \frac{(500)(10{,}000)}{10{,}000} + \frac{(0.25)(3)(10{,}000)}{2} = 30{,}000 + 500 + 3{,}750$$
$$= \$34{,}250/\text{year}$$

Hence, the optimal order quantity is 10,000 units paying $3/unit.

2. **[4 points]** Consider a product for which the fixed ordering cost is $500 per order. The annual demand for the product is relatively level and is 10,000 units per year. The system uses an annual inventory carrying rate of 25 percent. Suppose that an <u>incremental scheme</u> is effect with c_0=5, c_1=4.5, and c_2=4 with breakpoints of b_0=0, b_1=3,000, and b_2=6,000.
   - **a.** What are V_0, V_1, and V_2values for each price range?
   - **b.** What are the EOQ values for each price range? *Recall that the EOQ formula for incremental discounts is different from the standard EOQ formula.*
   - **c.** What are the candidate solutions from each price range?
   - **d.** What is the overall optimal solution?

SOLUTION:

a. We have

$$V_0 = \$0$$

$$V_1 = (5)(3{,}000 - 0) = \$15{,}000$$

$$V_2 = (5)(3{,}000 - 0) + (4.5)(6{,}000 - 3{,}000) = \$28{,}500$$

b. We also have

$$q_0^{EOQ} = \sqrt{\frac{2\lambda(K + V_0 - b_0c_0)}{ic_0}} = \sqrt{\frac{2(10{,}000)(500 + 0 - (0)(5))}{(0.25)(5)}} = 2{,}828.43 \text{ units}$$

$$q_1^{EOQ} = \sqrt{\frac{2\lambda(K + V_1 - b_1c_1)}{ic_1}} = \sqrt{\frac{2(10{,}000)(500 + 15{,}000 - (3{,}000)(4.5))}{(0.25)(4.5)}} = 5{,}962.85 \text{ units}$$

$$q_2^{EOQ} = \sqrt{\frac{2\lambda(K + V_2 - b_2c_2)}{ic_2}} = \sqrt{\frac{2(10{,}000)(500 + 28{,}500 - (6{,}000)(4))}{(0.25)(4)}} = 10{,}000 \text{ units}$$

c. Candidate solutions can be determined as follows:

$q_0^{EOQ} = 2{,}828.43$ units; Compare against the range $0 \le q < 3{,}000$;
Case 2 applies (between the lower and upper bounds);
$q_0^* = 2{,}828.43$ units.

$q_1^{EOQ} = 5{,}962.85$ units; Compare against the range $3{,}000 \le q < 6{,}000$;
Case 2 applies (between the lower and upper bounds);
$q_1^* = 5{,}962.85$ units.

$q_2^{EOQ} = 10{,}000$ units; Compare against the range $6{,}000 \le q < \infty$;
Case 2 applies (between the lower and upper bounds);
$q_2^* = 10{,}000$ units

d. There are 3 candidate solutions. Consequently, the overall optimal solution is given by:

$$C(2{,}828.43{,}5) = \frac{(500)(10{,}000)}{2{,}828.43} + \left(\frac{0.25}{2} + \frac{10{,}000}{2{,}828.43}\right)[0 + (2{,}838.43 - 0)(5)]$$
$$= 1{,}767.77 + 51{,}767.77 = \$53{,}535.53/\text{year}$$

$$C(5{,}962.85{,}4.5) = \frac{(500)(10{,}000)}{5{,}962.85} + \left(\frac{0.25}{2} + \frac{10{,}000}{5{,}962.85}\right)[15{,}000 + (5{,}962.85 - 3{,}000)(4.5)]$$
$$= 839.53 + 51{,}057.18 = \$51{,}895.70/\text{year}$$

$$C(10{,}000{,}4) = \frac{(500)(10{,}000)}{10{,}000} + \left(\frac{0.25}{2} + \frac{10{,}000}{10{,}000}\right)[28{,}500 + (10{,}000 - 6{,}000)(4)]$$
$$= 500 + 50{,}062.5 = 50{,}562.5\$/\text{year}$$

Hence, the optimal order quantity is 10,000 units at a unit cost of $4/unit.